

LAB 1

- 1.) Successful output of ip addr to show your statically set IP information is correct
(15 Points)

```
pulkit@desktop:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:50:56:86:81:87 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 42.49.30.201/24 brd 42.49.30.255 scope global ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::250:56ff:fe86:8187/64 scope link
```

- 2.) Successful output of ping to your default gateway
(15 Points)

```
pulkit@desktop:~$ ping 42.49.30.254
PING 42.49.30.254 (42.49.30.254) 56(84) bytes of data.
64 bytes from 42.49.30.254: icmp_seq=1 ttl=63 time=0.163 ms
64 bytes from 42.49.30.254: icmp_seq=2 ttl=63 time=0.197 ms
64 bytes from 42.49.30.254: icmp_seq=3 ttl=63 time=0.159 ms
64 bytes from 42.49.30.254: icmp_seq=4 ttl=63 time=0.133 ms
64 bytes from 42.49.30.254: icmp_seq=5 ttl=63 time=0.118 ms
64 bytes from 42.49.30.254: icmp_seq=6 ttl=63 time=0.161 ms
64 bytes from 42.49.30.254: icmp_seq=7 ttl=63 time=0.122 ms
64 bytes from 42.49.30.254: icmp_seq=8 ttl=63 time=0.178 ms
64 bytes from 42.49.30.254: icmp_seq=9 ttl=63 time=0.155 ms
64 bytes from 42.49.30.254: icmp_seq=10 ttl=63 time=0.162 ms
64 bytes from 42.49.30.254: icmp_seq=11 ttl=63 time=0.128 ms
64 bytes from 42.49.30.254: icmp_seq=12 ttl=63 time=0.171 ms
64 bytes from 42.49.30.254: icmp_seq=13 ttl=63 time=0.153 ms
64 bytes from 42.49.30.254: icmp_seq=14 ttl=63 time=0.219 ms
64 bytes from 42.49.30.254: icmp_seq=15 ttl=63 time=0.164 ms
64 bytes from 42.49.30.254: icmp_seq=16 ttl=63 time=0.167 ms
^C
--- 42.49.30.254 ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15342ms
rtt min/avg/max/mdev = 0.118/0.159/0.219/0.025 ms
pulkit@desktop:~$
```

3.) Screenshot of Iowa State's homepage successfully loading in firefox
(10 Points)



4.)

File	Why is editing my file necessary when setting up my network settings?
/etc/netplan/01-netcfg.yaml (20 points)	Editing this file is necessary for setting up my network because editing it allows us to set up a static IP address, the gateway and configure .
/etc/environment (20 points)	It is a system-wide configuration profile that sets the path variables for all the users on the system to a default value.
/etc/profile (20 points)	It sets up the system wide variables (such as PATH, UMASK etc.) to ensure security for all users.